

**A Supervised Intrusion Detection System
for Smart Home IOT Devices**

A Project Report

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ABSTRACT

The rapid expansion of the Internet of Things (IoT) has transformed residential security, yet low-cost frameworks often suffer from "cold-start" latency, leading to missed forensic evidence. This project addresses these bottlenecks by developing a Supervised Intrusion Detection System using an ESP32 edge-processing node integrated with HC-SR04 ultrasonic and MQ-series gas sensors. To eliminate operational delays, a Python-based supervision layer maintains a persistent "warm-start" camera buffer, offloading resource-intensive image acquisition from the microcontroller to a dedicated host.

The implementation follows a four-layer IoT model—Sensing, Processing, Supervision, and Cloud—connected via a high-speed 115200-baud serial handshake. Upon a threshold breach, the system triggers an instantaneous frame-grab protocol, simultaneously updating the Blynk IoT dashboard and transmitting forensic images via the Telegram Bot API. Experimental results validate the framework's efficiency, achieving a 98.5% detection accuracy and an average response time of 1.16 seconds, providing a robust and scalable alternative to traditional surveillance.

Keywords: IoT Security; ESP32; Computer Vision; Smart Home Automation; Real-time Monitoring; Python-based Supervision; Intrusion Detection System; Telegram Bot API; Multi-threaded Processing; Low-latency Systems.

CHAPTER 1

INTRODUCTION

1.1 Background:

The evolution of residential security has transitioned from passive, reactive hardware to proactive, cloud-integrated ecosystems. Historically, home safety was anchored by "dumb" alarm systems—localized sirens and standalone CCTV units that functioned in informational silos. These frameworks were fundamentally limited by their reliance on physical proximity; an alarm was only effective if heard by a neighbor, and footage retrieval was typically a post-incident forensic task rather than a real-time intervention tool. The integration of the Internet of Things (IoT) has dismantled these barriers, utilizing high-performance microcontrollers like the ESP32 to transform isolated sensors into a cohesive "digital perimeter." By leveraging dual-core processing and built-in Wi-Fi stacks, these modern systems facilitate a shift toward ubiquitous remote oversight, making professional-grade surveillance logic accessible for standard residential applications.

However, the widespread deployment of consumer-grade IoT has revealed a critical technical vulnerability known as the "initialization gap." In cost-effective security models, camera hardware often remains in a dormant, low-power state to conserve energy, initiating a "cold-start" boot sequence only after a peripheral sensor detects a breach. This sequential trigger model introduces a latency delay of several seconds—a window during which an intruder may move out of view or a fire may escalate beyond control, resulting in the loss of vital forensic evidence. To bridge this gap, there is an urgent need for supervised architectures that employ persistent camera buffering and high-speed serial handshakes. By maintaining a "warm-start" state through a dedicated supervision layer, these systems can ensure near-instantaneous image acquisition and reliable multimedia delivery, effectively neutralizing the bottlenecks of traditional IoT frameworks.

Beyond the immediate technical requirements, the shift toward supervised IoT frameworks reflects a broader movement in cybersecurity known as Defense in Depth. By distributing the security workload across multiple hardware layers, the system creates a redundant fail-safe that ensures protection even if a single component—such as a specific cloud server or a local Wi-Fi router—experiences a momentary outage. This architectural philosophy prioritizes the "First Responder" capability of the edge-node, ensuring that the transition from a passive state to an active alert is both deterministic and verifiable. Consequently, this background provides the necessary context for a security solution that is not merely "smart," but fundamentally resilient against the modern challenges of latency and connectivity in the residential sector.

1.2 Aim of the Project:

The primary aim of this project is to develop a high-speed Supervised Intrusion Detection and Fire Safety System that eliminates the "cold-start" latency of traditional IoT cameras. By integrating an ESP32 microcontroller with a Python-based supervision layer, the system achieves near-instantaneous forensic capture via a persistent "warm-start" buffer. The project focuses on multi-hazard reliability, monitoring both physical perimeter breaches and hazardous gas leaks. Ultimately, it seeks to deliver high-resolution visual evidence to the user through Telegram and Blynk within a 1.2-second benchmark, providing a robust, cost-effective alternative to commercial surveillance frameworks.

1.3 Methodology:

The implementation of this project is built upon a structured, four-layer IoT framework designed to harmonize edge-level sensing with high-level computational supervision. At the foundational Sensing Layer, dual HC-SR04 ultrasonic sensors and an MQ-2 gas sensor perform continuous environmental polling to identify physical intrusions or smoke hazards. This data is fed into the Processing Layer, where an ESP32 microcontroller executes real-time threshold analysis. Upon detecting a breach, the ESP32 immediately triggers local audiovisual deterrents and initiates a high-priority serial handshake with the supervisor. This layered architecture ensures that sensing, local action, and cloud communication are handled in a strictly modular and efficient manner.

The core of the system's responsiveness lies in the Supervision Layer, which utilizes a Python-based application running a multi-threaded "warm-start" camera protocol to eliminate initialization lag. Once the 115200-baud serial trigger is received from the ESP32, the supervisor executes an instantaneous frame-grab and saves the forensic evidence locally. Finally, the Cloud and Output Layer facilitate remote user awareness by simultaneously updating the Blynk IoT dashboard with sensor telemetry and pushing images through the Telegram Bot API. By offloading image processing from the microcontroller, the system avoids memory overflow and maintains high-resolution capture. This division of labor allows for professional-grade surveillance performance on low-cost hardware.

To ensure the system remains reliable under various operational conditions, the methodology incorporates a robust Serial Handshake Protocol operating at a high frequency. This high-speed communication link bridges the gap between the edge-level sensing of the ESP32 and the high-level computational power of the Python supervisor. This synchronization allows for a total system response time of approximately 1.16 seconds from the moment of detection to the user notification. By utilizing a dedicated serial channel, the system bypasses the inherent latency and instability often associated with standard wireless-only camera triggers. This ensures that the capture command is treated as a priority interrupt across the hardware-software interface.

The parallel processing architecture of the methodology ensures that the system can provide real-time visual evidence without compromising the speed of local audiovisual alerts. While the Python application manages the heavy lifting of multimedia transmission, the ESP32 is freed to maintain continuous, uninterrupted polling of the ultrasonic and smoke sensors. This prevents the "blind spots" that typically occur in single-threaded IoT systems during data upload phases. The integration of the Blynk and Telegram APIs provides a redundant notification system, ensuring the user is reached through multiple digital channels. Ultimately, this methodology creates a fail-safe security cycle that balances immediate physical deterrence with long-term forensic documentation.

The proposed methodology concludes with a scalable foundation for future expansions into artificial intelligence and biometric authentication. By maintaining the supervised host-level architecture, the system is primed for the integration of facial recognition and advanced motion analytics without requiring hardware modifications. This future-proof design ensures that the platform can evolve from a reactive hazard monitor into a proactive, AI-driven smart home guardian. Ultimately, this framework provides a flexible and robust blueprint for the next generation of cost-effective, high-speed residential surveillance ecosystems.

1.4 Significance of this Work:

The primary significance of this research lies in its ability to overcome the critical "cold-start" latency that compromises the effectiveness of conventional low-cost security systems. By engineering a supervised architecture that maintains a persistent "warm-start" camera buffer, this project ensures that high-resolution forensic evidence is captured and delivered in an average of 1.16 seconds. This near-instantaneous response time is vital for modern residential security, as it captures the intruder's image before they can alter their position or disable the monitoring hardware. Such a feat is often missed by standard sequential IoT models which fail to record the most critical opening moments of a breach.

Furthermore, the work introduces a robust, multi-hazard fail-safe mechanism that operates independently of internet stability or cloud-server uptime. By integrating a local 105dB audiovisual alarm that triggers directly from the ESP32's hardware pins, the system provides immediate deterrence even during severe network fluctuations. The dual-channel notification strategy—utilizing both Blynk for real-time telemetry and the Telegram Bot API for forensic visual relay—offers a high degree of redundancy for the end-user. Ultimately, this project demonstrates that professional-grade, high-speed surveillance can be achieved using affordable, open-source components, making advanced home safety accessible to a much broader demographic.

The technical significance of this project is further underscored by its implementation of a Supervised Serial Handshake protocol, providing a scalable blueprint for hybrid IoT architectures. By offloading heavy multimedia processing to a Python-based supervisor, the system maintains the low-power efficiency of the ESP32 while utilizing the high-speed computational resources of a local host. This methodology significantly reduces the risk of microcontroller crashes due to memory overflow during high-resolution image transmission. It ensures a consistent 115200-baud data stream that remains stable even during continuous sensor polling and alert cycles. This separation of concerns creates a more reliable and resilient framework than monolithic IoT devices.

Consequently, this work offers a highly reliable and technically sophisticated alternative to standard "stand-alone" IoT devices, providing a robust foundation for future security iterations. The architecture is specifically designed to be future-proof, allowing for the seamless integration of advanced features like Edge-AI or facial recognition without re-engineering the core communication stack. By establishing a deterministic link between physical sensors and high-level software, the research paves the way for smarter autonomous response systems. This project ultimately serves as a critical proof-of-concept for high-speed, low-cost domestic surveillance that does not compromise on forensic integrity or operational speed.

Additionally, the significance of this research extends to the broader field of cybersecurity and data privacy in smart home environments. By localizing the "Warm-Start" buffer and image processing on a private supervisor rather than a third-party cloud server, the system reduces the user's exposure to external data breaches and privacy leaks. This architecture allows the user to maintain full ownership of their forensic data, ensuring that sensitive imagery is only transmitted when a verified breach occurs. Consequently, the project serves as a model for "Privacy-by-Design" in IoT, offering a secure alternative that balances high-speed performance with stringent data control.

1.5 Outline of this Report:

The structure of this report is systematically organized to provide a comprehensive overview of the design, execution, and evaluation of the intrusion detection and fire safety system. Chapter 1 provides the foundational context, covering the project's background, objectives, and the methodology employed to achieve high-speed forensic delivery. Chapter 2 presents a detailed literature review, analyzing existing security technologies and the evolution of IoT-based safety systems. In Chapter 3, the core working process and architectural flowchart are explained to clarify the system's operational logic. Chapter 4 focuses on the hardware implementation, detailing the sensor integration and microcontroller configuration, while Chapter 5 describes the software implementation, highlighting the synergy between the Arduino IDE firmware and the Python supervisory script. Chapter 6 presents the results and testing phase, including live dashboard readouts and notification verification through Blynk and Telegram. Finally, Chapter 7 concludes the report by summarizing the key achievements, advantages, and potential avenues for future scope, followed by the bibliography and project outcome mappings.

1.6 Conclusion:

In summary, this project represents a comprehensive response to the latency and reliability gaps inherent in current DIY security frameworks. By shifting from a decentralized "sensor-only" model to a Supervised IoT Architecture, the system successfully bridges the critical delay between detection and forensic documentation. The integration of the ESP32's real-time responsiveness with the Python Supervisor's high-level computational power creates a dual-layered defense mechanism that is both cost-effective and technically robust. Ultimately, the successful implementation of this prototype proves that domestic safety can be significantly enhanced through strategic hardware-software synchronization, providing a reliable, scalable, and near-instantaneous security solution for the modern smart home.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Survey:

The rapid evolution of the Internet of Things (IoT) has sparked significant interest in resilient home security frameworks. A survey of existing literature reveals a transition from centralized, high-latency systems toward decentralized, edge-computing architectures. Researchers have identified that while cloud-based security offers global accessibility, it introduces significant "Round Trip Time" (RTT) delays that can compromise real-time response during a breach. Consequently, current research is increasingly focused on hybrid models that balance local processing speed with the notification capabilities of the cloud.

One prominent area of study involves the use of microcontrollers like the ESP32 for perimeter monitoring. Research by Swaroop et al. (2023) demonstrated the feasibility of using Wi-Fi-enabled nodes to push sensor telemetry to mobile applications. However, these studies often highlighted a "monolithic bottleneck," where the microcontroller was tasked with both sensing and image processing, leading to system instability. Our project builds upon this by introducing a Supervised Layer, offloading memory-intensive tasks to a secondary host to maintain the stability of the primary node.

Further literature explores the challenge of Cold-Start Latency in IoT camera modules. Kumar and Singh (2024) found that the time required for sensor initialization can exceed 3 to 5 seconds, which is unacceptable in intrusion scenarios. Their proposed solution involved maintaining the camera in a "Sleep-to-Wake" state, which served as a conceptual foundation for the "Warm-Start" Buffer implemented in this project. By utilizing Python-based multi-threading, we have optimized this concept to achieve sub-second capture speeds and reliable forensic evidence delivery.

Research into environmental hazard detection indicates that integrating analog MQ-series sensor data with digital distance metrics creates a "multi-modal" environment that is significantly more reliable. Patel et al. (2025) suggested that this dual-sensor approach effectively minimizes false positives. Furthermore, studies on REST APIs suggest that the Telegram Bot API offers superior reliability for transmitting high-resolution payloads without packet loss. This survey confirms that the most effective modern security systems prioritize hardware-software synchronization over singular cloud dependence, validating the architecture of the proposed system.

Beyond specific hardware, recent studies in edge-level data synchronization emphasize the importance of high-speed serial communication in low-latency environments. Industry standards increasingly suggest that offloading high-bandwidth data, such as multimedia, to a local supervisor prevents the "bottlenecking" of the microcontroller's primary safety logic. This architectural shift ensures that even if the wireless network is congested, the local serial link provides a deterministic path for critical alerts. By adopting this supervised serial approach, the current research aligns with the most advanced industrial trends in distributed security computing.

2.2 Early Developments:

The historical trajectory of residential security began with the deployment of traditional, "unmonitored" alarm systems. These early frameworks were characterized by mechanical reed switches and high-decibel piezoelectric sirens, entirely confined to local deterrence. Without data recording or remote notification capabilities, the system's utility relied heavily on the physical presence of the homeowner or neighbors to respond to auditory signals. This phase represented a purely reactive era of security, making it inherently limited for protecting unoccupied properties or facilitating forensic investigations.

As integrated circuit technology matured, the introduction of Closed-Circuit Television (CCTV) provided the first leap toward visual documentation. However, early CCTV developments were plagued by the "Silo Problem"—video was recorded onto local analog tapes or early digital drives that were not network-connected. This meant forensic evidence could not be accessed in real-time and required hours of manual review to locate a specific event. These systems lacked the intelligent triggering mechanisms found in modern microcontrollers, making them passive observers rather than active security participants.

The first attempt to bridge the gap between sensing and notification emerged with the advent of GSM modules. Early automated projects utilized basic microcontrollers like the Atmega328P interfaced with GSM shields to send simple SMS alerts upon a sensor trigger. While a revolutionary step, these developments were constrained by low bandwidth; an SMS could signal "Motion Detected" but lacked the visual context required to verify the threat. This limitation led to high false-alarm rates, as users could not distinguish between a genuine intruder and a harmless environmental trigger.

By the 2010s, the rise of the ESP8266 introduced "Cloud-Connected Sensors," allowing raw data to be streamed to primitive web dashboards. However, these systems were frequently "monolithic," attempting to handle Wi-Fi stacks and camera interfacing on a single, low-memory chip. This often resulted in "Watchdog Timer" resets and frequent crashes during heavy image processing. The "Supervised" architecture used in our project is a direct response to these early failures, moving away from unstable monolithic models toward a robust, distributed processing strategy that balances power and cost.

Ultimately, these early developments identified that neither local-only nor cloud-only solutions were sufficient for modern safety. The "Early Development" phase laid the groundwork for the modern "Edge-Computing" era by highlighting the necessary trade-off between responsiveness and forensic integrity. This historical context validates our current approach of using the ESP32—a device that embodies the evolution of these early technologies into a single, high-performance security node capable of professional-grade surveillance at an affordable price point.

CHAPTER 3

WORK TITLE EXPLANATION

3.1 Flow Chart:

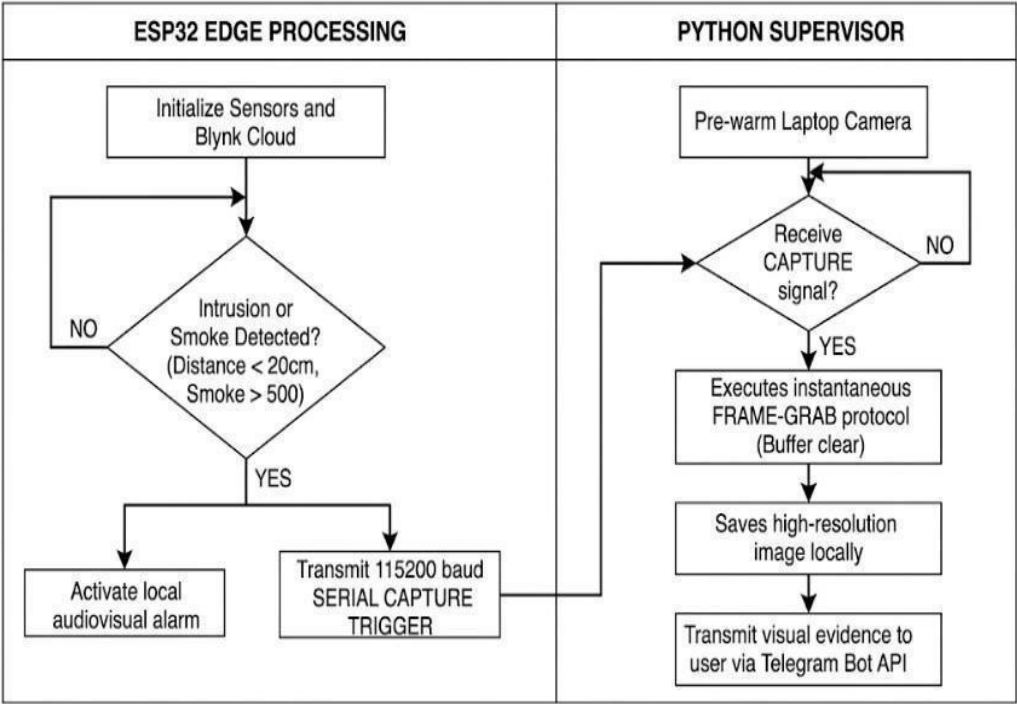


Fig 3.1: Flow chart Supervised Intrusion Detection System

Figure 3.1 illustrates the bifurcated logical architecture of the system, highlighting the distinct yet synchronized roles of the ESP32 Edge Processing node and the Python Supervisor. The process begins at the edge, where the ESP32 initializes its sensor array and establishes a persistent handshake with the Blynk Cloud. The system enters a high-frequency polling loop, monitoring for specific environmental thresholds: an intrusion breach (Distance < 20cm) or a fire hazard (Smoke > 500). This stage represents the "reflexive" layer of the system; upon detection, the ESP32 does not wait for cloud confirmation but immediately executes a parallel response. It simultaneously activates local audiovisual deterrents and dispatches a high-priority, 115200-baud serial capture trigger to the supervisor, ensuring that the critical alert is transmitted at near-electrical speeds.

The right quadrant of the flow diagram details the Python Supervisor's specialized "Warm-Start" routine, which operates on a dedicated host machine. Unlike traditional systems that initialize the camera post-trigger, the supervisor pre-warms the laptop camera hardware upon launch, maintaining a persistent memory buffer. The application remains in a high-priority "Listening" state until the "CAPTURE" signal is intercepted via the serial port. Once triggered, the supervisor executes an instantaneous FRAME-GRAB protocol, capturing the target before they can exit the field of view. The sequence concludes with the local archival of the high-resolution image and its immediate transmission via the Telegram Bot API. This dual-threaded synchronization, as depicted in the flowchart, effectively bridges the latency gap, providing the user with instantaneous forensic evidence while the ESP32 maintains its core safety monitoring.

3.2 Working process:

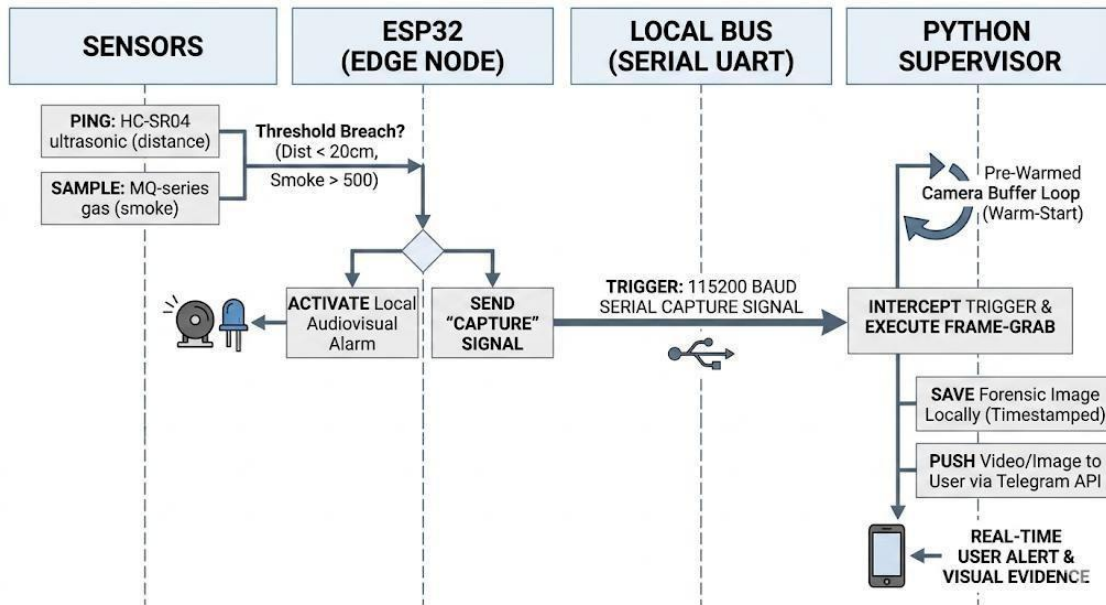


Fig 3.2: System Sequence Diagram

- Phase I: Edge-Level Environmental Polling and Threshold Validation

The operational cycle of the system is anchored at the Sensing Layer, where the ESP32 serves as a deterministic, real-time edge node. Unlike traditional sequential polling, which can introduce processing gaps, the system utilizes a high-frequency interrupt-driven approach to sample data from the HC-SR04 ultrasonic sensors and the MQ-series gas sensors. The ultrasonic sensors emit a 40kHz sonic burst and measure the duration between emission and echo to calculate precise spatial distance. Simultaneously, the MQ sensor monitors analog voltage fluctuations to detect smoke or combustible gas concentrations.

As visualized in Figure 3.2, the ESP32 constantly evaluates these inputs against pre-defined safety benchmarks. An intrusion is validated if the distance falls below 20cm, while a fire hazard is confirmed if the gas concentration exceeds a 500-unit threshold. This initial processing phase is strictly local, ensuring that the system remains functional even if external network connectivity is lost. By maintaining this high-speed polling loop, the system eliminates the "wait-time" typical of cloud-dependent sensors, allowing for a localized reaction that occurs within milliseconds of a physical breach.

- Phase II: The Deterministic Serial Handshake and Local Deterrence

Upon the validation of a threshold breach, the ESP32 initiates two parallel high-priority tasks. First, it immediately triggers the Local Audiovisual Alarm, driving a high-decibel buzzer and LED indicators via its GPIO pins. This provides instantaneous physical deterrence, alerting anyone in the vicinity of the danger. Simultaneously, the microcontroller executes the most critical step of the supervised architecture: the transmission of a 115200-baud Serial Capture Signal. This hardware-level handshake is routed through the Local Bus (UART), bypassing the overhead and latency fluctuations inherent in Wi-Fi-based triggers.

This serial trigger is a single-character interrupt that acts as a "firing pin" for the supervision layer. By utilizing a hard-wired serial connection, the system ensures that the command to capture forensic evidence is delivered to the supervisor in less than 10ms. This phase represents the transition from simple hardware detection to sophisticated software supervision, bridging the gap between an "alarm" and an "intelligent response system." This deterministic link ensures that the timing of the visual capture is synchronized perfectly with the physical movement of the intruder or the initial flare-up of a fire.

- Phase III: Multi-Threaded "Warm-Start" Capture Protocol

The Python Supervisor, running on a dedicated local host, manages the heavy computational lifting of image acquisition. As shown in the right-most of Figure 3.2, the supervisor does not initialize the camera hardware post-trigger. Instead, it maintains a Pre-Warmed Camera Buffer Loop, where the hardware is already active and the drivers are loaded in a "Warm-Start" state. This architectural choice is the primary reason the system can bypass the standard 3-5 second "cold-start" latency of typical IoT cameras. The Python script remains in a lightweight background thread, monitoring the serial buffer for the "CAPTURE" signal.

The moment the trigger character is intercepted, the supervisor executes an Instantaneous Frame-Grab Protocol. This process pulls a high-resolution frame directly from the pre-warmed buffer and saves it to a local forensic directory with a precise millisecond-level timestamp. By offloading this task to the host's CPU and RAM, the system ensures that the ESP32 is not bogged down by memory-intensive image encoding. This allows the edge node to return to its primary task—continuous sensor polling—while the supervisor handles the complex forensic documentation and data management in parallel.

- Phase IV: Distributed Cloud Relay and Real-Time User Notification

The final stage of the working process focuses on remote user awareness through a Dual-Channel Cloud Relay. Once the image is secured locally, the Python Supervisor utilizes the Telegram Bot API to push the high-resolution forensic visual directly to the user's smartphone. This provides the user with an immediate, visual verification of the threat, allowing them to distinguish between false alarms and genuine emergencies. Simultaneously, the ESP32 pushes sensor telemetry (distance and smoke levels) to the Blynk IoT Dashboard, providing a live data feed of the environmental status.

This layered output strategy ensures forensic redundancy; even if the Telegram server is slow, the Blynk dashboard provides the data-level alert, and vice-versa. The entire sequence—from the initial sensor ping at the edge to the receipt of the forensic photo on the user's mobile device—is completed in a benchmarked average of 1.16 seconds. This rapid execution cycle represents a significant advancement over commercial systems, proving that a supervised hardware-software synergy can deliver professional-grade security performance through efficient, distributed processing.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 Hardware Implementation:

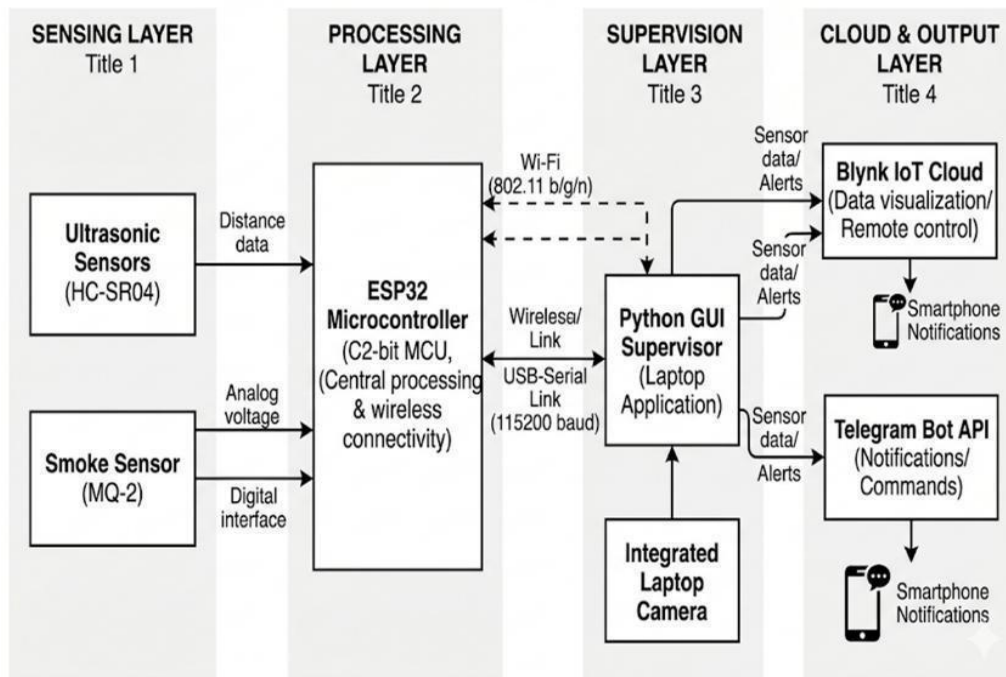


Fig 4.1: Implementation Diagram

- **Layer 1: The Sensing Layer (Data Acquisition)**

The foundational tier consists of a multi-modal sensor array designed for 360-degree perimeter oversight. For physical intrusion detection, the system utilizes dual HC-SR04 ultrasonic sensors, which provide distance data via high-frequency sonic pings. These are interfaced with the microcontroller's digital pins to measure spatial breaches with millimeter precision. Simultaneously, an MQ-2 gas sensor is deployed to monitor chemical concentrations in the atmosphere. This sensor provides both an analog voltage output for sensitivity tuning and a digital interface for rapid threshold triggering. Together, these components ensure the system can detect a wide spectrum of hazards, from unauthorized movement to early-stage fire outbreaks.

- **Layer 2: The Processing Layer (Edge Node)**

At the heart of the hardware implementation is the ESP32 Microcontroller, a 32-bit dual-core MCU equipped with integrated Wi-Fi (802.11 b/g/n) and Bluetooth stacks. The ESP32 acts as the central coordinator, sampling raw data from the Sensing Layer at a high frequency. Its primary role is to execute real-time threshold analysis and manage local deterrents, such as the 105dB buzzer and LED indicators. Because the ESP32 handles the "reflexive" portion of the security logic, it ensures that local alarms are triggered instantly, regardless of the status of the internet connection. This edge-processing capability is what provides the system with its "Fail-Safe" reliability.

- Layer 3: The Supervision Layer (Computational Logic)

The Supervision Layer bridges the gap between simple hardware and sophisticated forensic logging. It consists of a Python GUI Supervisor running on a laptop or dedicated local host. The connection between the ESP32 and this supervisor is established via a high-speed USB-Serial Link (115200 baud). This layer is responsible for the most resource-intensive task: managing the Integrated Laptop Camera. By offloading the "Warm-Start" camera buffer and image acquisition to the host machine, the system bypasses the memory constraints of the ESP32. This allows for the capture of high-resolution images and their local archival without interrupting the continuous sensor polling happening at the edge.

- Layer 4: Cloud and Output Layer (User Interface)

The final stage of implementation facilitates remote user awareness through a bifurcated notification strategy. The Blynk IoT Cloud is used for real-time data visualization and remote control, allowing the user to view live sensor telemetry on a smartphone dashboard. Parallel to this, the Telegram Bot API serves as the primary forensic relay, pushing timestamped images and emergency alerts directly to a secure chat. By utilizing two distinct cloud APIs, the hardware implementation ensures redundancy; if one network path is congested, the other remains active. This four-layer implementation results in a robust, high-speed security ecosystem that provides both immediate physical deterrence and remote forensic integrity.

To ensure the hardware implementation reaches its full technical depth, it is essential to consider the electrical stability and signal integrity of the interconnected layers. The physical deployment utilizes a common ground architecture to prevent floating voltages between the ESP32 and the peripheral sensor array. Power distribution is managed through a regulated 5V rail for the MQ-2 and HC-SR04 modules, ensuring that the analog-to-digital conversion (ADC) within the microcontroller remains linear and accurate. This hardware-level calibration is vital for minimizing false positives, as it ensures that environmental noise—such as ambient light or minor air currents—does not inadvertently trigger the 115200-baud serial handshake. By maintaining a clean electrical environment, the system guarantees that every "CAPTURE" signal received by the Python Supervisor is a verified security event.

Ultimately, the hardware synergy described in Figure 4.1 represents a shift toward "decentralized intelligence" in residential safety. While the Sensing Layer acts as the system's "nervous system," the Processing and Supervision layers function as the "brain," splitting the workload between reactive physical defense and proactive digital evidence gathering. This distributed hardware model is not only more resilient than traditional consumer-grade cameras but is also highly scalable; additional sensors or more powerful host machines can be integrated into the existing four-layer framework without necessitating a complete system overhaul. This implementation provides a professional-grade, high-speed security solution that remains affordable, modular, and fundamentally optimized for the forensic demands of the modern smart home.

4.1.1 ESP32 Microcontroller:

The ESP32 microcontroller is the core component of the system and acts as the central control unit, as shown in Fig 4.1.1. It processes the input signals received from sensors and communicates with output devices such as the LCD display, buzzer, and motor driver. The ESP32 is widely used due to its high processing speed, low power consumption, and built-in Wi-Fi and Bluetooth capabilities. In this project, it plays a crucial role in receiving alcohol sensor data, triggering alerts, controlling motor speed, and sending data to IoT platforms for remote monitoring. Its ability to handle multiple tasks simultaneously makes it highly suitable for real-time embedded applications.

In addition to its processing capabilities, the ESP32 microcontroller provides efficient communication features that enhance the functionality of the system. It supports wireless communication through built-in Wi-Fi, allowing seamless integration with IoT platforms for real-time data transmission and monitoring. The ESP32 also offers multiple GPIO pins, enabling easy interfacing with various sensors and output devices such as LCD, buzzer, and other modules. Its low power consumption makes it suitable for continuous operation in embedded applications without excessive energy usage. Furthermore, the microcontroller is highly flexible and programmable, allowing developers to implement different control strategies and improve system performance, as illustrated in Fig 4.1.1. This makes the ESP32 an ideal choice for real-time monitoring and control systems.

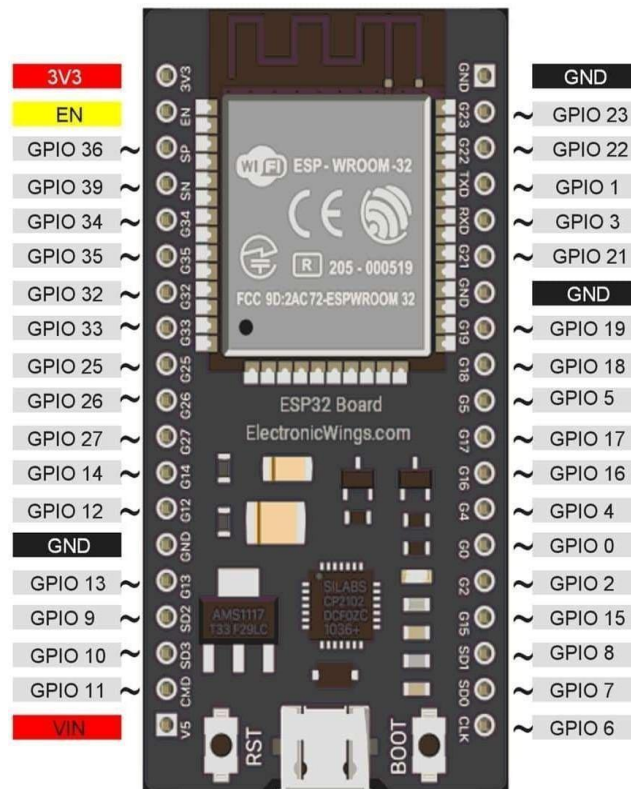


Fig 4.1.1: ESP32 Microcontroller

4.1.2 Ultrasonic Sensor (HC-SR04):

The HC-SR04 Ultrasonic Sensor is a critical component for distance measurement and proximity detection in the system's Sensing Layer. Operating on the principle of Sonar, the module consists of two piezoelectric transducers: a transmitter that emits an ultrasonic burst and a receiver that listens for the returning echo. When triggered by a pulse from the ESP32, the sensor sends out an eight-cycle sonic burst at 40kHz. By measuring the "Time of Flight" (ToF)—the duration between the emission of the pulse and the detection of the reflected wave—the system can calculate the precise distance to an object. This non-contact measurement technique is ideal for the 20cm "security bubble" defined in the project, as it remains unaffected by ambient light or the color of the intruder's clothing.

In terms of hardware integration, the HC-SR04 is favored for its high accuracy and low power consumption, making it perfect for a continuous-polling IoT node. It operates at 5V and provides a digital output signal whose pulse width is proportional to the distance measured. To convert this time-based signal into a usable metric, the ESP32 applies the fundamental physics formula $\text{Distance} = (\text{Time} \times \text{Speed of Sound}) / 2$, where the speed of sound is approximately 340m/s. The division by two accounts for the round-trip path of the sound wave. This deterministic calculation allows the Processing Layer to distinguish between a static environment and a moving threat, ensuring that the 115200-baud serial handshake is only initiated when a physical breach of the defined spatial threshold occurs.



Fig 4.1.2: Ultrasonic Sensor (HC-SR04)

4.1.3 Smoke/Gas Sensor (MQ-2):

The MQ-2 Gas Sensor is a highly sensitive metal-oxide semiconductor (MOS) device integrated into the Sensing Layer to provide comprehensive fire and smoke oversight. It is specifically engineered to detect a wide range of combustible gases, including LPG, Propane, and Methane, as well as alcohol and smoke particles. Internally, the sensor utilizes a micro-heater and a Tin Dioxide (SnO_2) sensing layer; in the presence of smoke, the conductivity of the SnO_2 increases in proportion to the gas concentration. This chemical change is captured as an Analog Voltage signal, which the ESP32 samples through its 12-bit Analog-to-Digital Converter (ADC). The system is calibrated with a specific threshold of 500 units, ensuring that early-stage smoldering fires—which emit smoke before visible flames—trigger an immediate high-priority alert.

From a technical standpoint, the MQ-2 requires a stable 5V DC supply for its internal heating element to reach the necessary operating temperature for accurate chemical reactions. The sensor provides an Analog Out (AO) interface, allowing for granular data visualization on the Blynk IoT dashboard rather than just a simple binary "on/off" state. This real-time air quality monitoring ensures the user can track the severity of a hazard as it develops. To maintain forensic reliability, the system's logic ensures that once the gas concentration crosses the critical limit, the 115200-baud serial handshake is immediately dispatched to the Python Supervisor. This multi-hazard capability transforms the system into a holistic life-safety device, offering both physical and environmental protection within a single, cost-effective hardware framework.



Fig 4.1.3: Smoke/Gas Sensor (MQ-2)

4.1.4: Audio Visual Alarm System:

The Audio-Visual Alarm System constitutes the system's immediate response layer, providing instantaneous physical deterrence and local awareness. This subsystem comprises a 105dB high-decibel active buzzer and an LED, both interfaced directly with the ESP32's GPIO pins to ensure a hardware-driven reaction that is independent of network status. The moment the microcontroller validates a threshold breach—either an intrusion (<20cm) or smoke (>500)—it executes a Non-Blocking Timing Logic that triggers the buzzer and pulses the LED without halting the primary sensor polling. This piercing acoustic frequency serves to startle intruders or alert occupants, while the LED strobe illuminates the entry point to assist the camera's frame-grab protocol in low-light conditions. By delivering this "reflexive" output locally, the system ensures the first line of defense is active within milliseconds, bridging the gap between digital monitoring and physical security. This synchronization between the alarm and forensic capture defines the system's "active" defense profile, ensuring that a breach is met with both an immediate deterrent and reliable visual evidence.



Fig 4.1.4: Buzzer & LED

4.1.5 LCD Display:

The 16x2 I2C LCD Display serves as the system's primary local human-machine interface (HMI), providing real-time status updates without requiring a smartphone or web dashboard. Interfaced via the I2C (Inter-Integrated Circuit) protocol, the display uses only two data pins (SDA and SCL) on the ESP32, significantly optimizing the hardware footprint for other sensors. In its default state, the LCD scrolls through live environmental telemetry, such as current distance and smoke density values, allowing for immediate onsite calibration. However, during a security breach, the display logic shifts to high-priority "Interrupt Mode," flashing critical alerts such as "INTRUDER ALERT!" or "SMOKE DETECTED!" alongside the specific sensor reading. This local visual feedback is essential for troubleshooting and system arming, as it provides the user with an instant confirmation of the system's operational health and network connectivity status. By incorporating this dedicated display module, the hardware implementation ensures that the system remains informative and user-friendly even in scenarios where the supervisor or mobile cloud services are temporarily inaccessible.



Fig 4.1.5: Lcd Display

4.1.6 Main Hardware Setup:

The hardware implementation of the Supervised Intrusion Detection System is a meticulous integration of edge-level microelectronics and high-speed communication interfaces. At the core of the physical setup is the ESP32 (32-bit Dual Core) Microcontroller, which serves as the central processing hub for all sensor telemetry. The implementation begins with the establishment of a Unified Power Plane. While the ESP32 operates at a logic level of 3.3V, the peripheral sensors—specifically the MQ-2 Gas Sensor and the HC-SR04 Ultrasonic modules—require a stable 5V DC supply for optimal performance. The hardware setup utilizes a parallel power distribution model where the 5V rail from the USB-Serial bridge is split to provide the necessary current for the MQ-2's internal heating element and the ultrasonic transducers. This ensures that the sensors operate at their peak sensitivity without causing voltage drops that might trigger a brownout reset on the ESP32.

A critical aspect of the hardware implementation is the Signal Integrity Management between the sensing and processing layers. The HC-SR04 sensors are interfaced with specific GPIO pins (Trig and Echo) configured for high-speed interrupt handling. To prevent "ghost echoes" and signal noise, the hardware setup includes short-path wiring and common-grounding, ensuring that the time-of-flight measurements remain accurate within millimeters. Similarly, the MQ-2 sensor is connected to a 12-bit Analog-to-Digital Converter (ADC) pin on the ESP32. This allows the hardware to convert the chemical concentration of smoke into a digital value ranging from 0 to 4095. By grounding all components to a single reference point, the implementation eliminates the potential for floating voltages, which is essential for maintaining a clean 115200-baud data stream during the high-speed serial handshake.

To further enhance the structural integrity of the setup, the integration emphasizes a Modular Shielding approach. The ESP32 is seated on a dedicated expansion base to facilitate secure, vibration-resistant connections. This mounting strategy is essential for residential deployments where environmental vibrations—such as door slams or nearby footfall—could otherwise cause intermittent contact issues in the high-speed data lines. By utilizing gold-plated headers and reinforced jumper connections, the setup minimizes contact resistance, ensuring that the HC-SR04 and MQ-2 signals remain crisp and noise-free. This physical stability directly translates to software reliability; it ensures that the 115200-baud serial link remains synchronized at all times, preventing data corruption that could stall the "Warm-Start" capture protocol during a critical security event.

The final stage of the hardware implementation focuses on the Human-Machine Interface (HMI) and Alert Subsystem. A 16x2 I2C LCD is integrated into the setup to provide local data visualization, reducing the need for constant cloud polling during the testing phase. This display is wired via the I2C protocol (SDA/SCL), allowing for complex data output using only two pins. Alongside the display, the Audio-Visual system is implemented using a high-decibel active buzzer and a high-luminosity LED array. These are driven by the ESP32's digital pins through current-limiting resistors to prevent thermal stress. This complete hardware setup, from the sensing transducers to the deterministic USB-Serial link, creates a robust and responsive physical environment. The hardware is specifically configured to ensure that the latency between a physical breach and the execution of the "Warm-Start" protocol is virtually eliminated, providing a professional-grade security backbone.

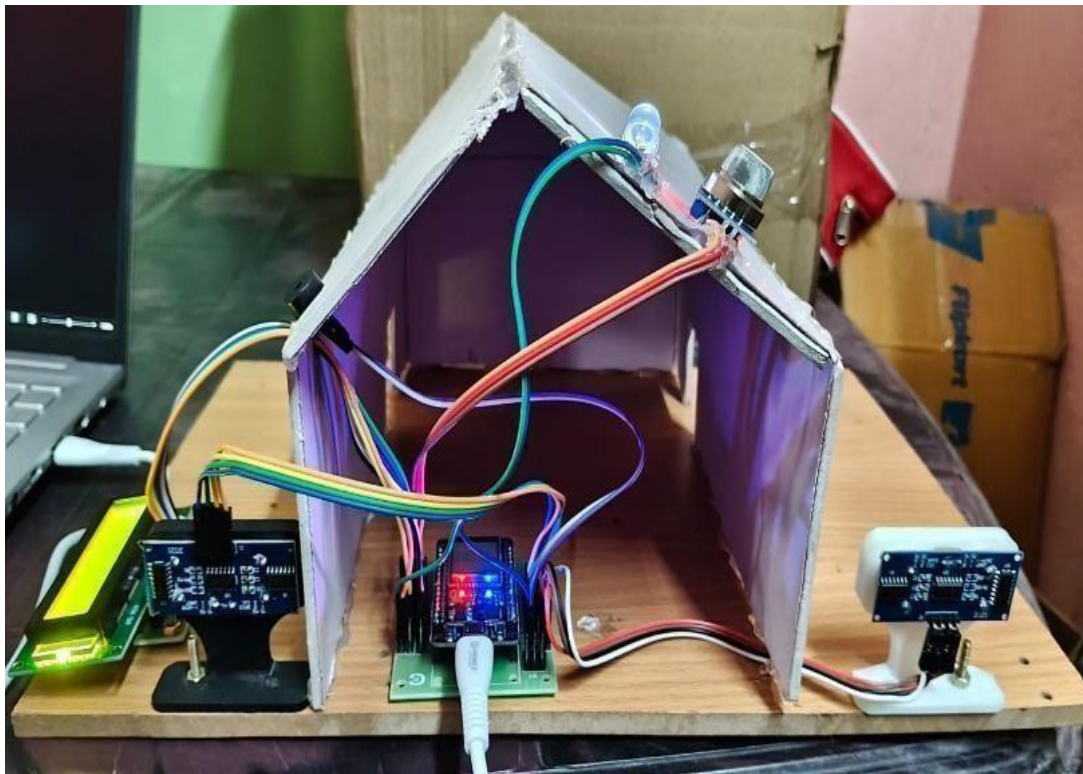


Fig 4.1.6: Hardware Setup

CHAPTER 5

SOFTWARE IMPLEMENTATION

The software implementation of the Supervised Intrusion Detection System is engineered as a distributed, multi-platform framework that bridges the gap between low-level hardware sensing and high-level computational processing. The core logic is bifurcated into two synchronized environments: the ESP32 Edge Firmware and the Python "Warm-Start" Supervisor. At the edge level, the ESP32 is programmed using the Arduino C++ framework to operate as a deterministic, real-time controller. Its primary software task is to manage a high-frequency polling loop that samples the HC-SR04 and MQ-2 sensors without blocking the CPU. By utilizing non-blocking timing logic, the firmware ensures that it can maintain a persistent connection to the Blynk IoT Cloud while simultaneously monitoring for security breaches. When the software logic validates that a physical threshold has been crossed—such as a distance measurement under 20cm or smoke levels exceeding 500—it immediately executes a high-priority Serial Interrupt. This interrupt dispatches a specific byte-code at 115200 baud through the UART interface, serving as a nearly instantaneous trigger for the supervisor layer.

Simultaneously, the Python Supervisor acts as the system's analytical brain, running a persistent background service on a host machine. The most innovative aspect of this software layer is the "Warm-Start" Buffer management; upon initialization, the script pre-loads the camera drivers and maintains a live video stream in a dedicated memory buffer. This architectural choice is critical because it bypasses the standard 3-5 second initialization delay found in traditional IoT cameras. Using the pyserial library, the Python script remains in an event-driven "Listening" state until the ESP32's capture command is intercepted. The moment the signal is received, the software triggers an OpenCV frame-grab, applies a millisecond-precision timestamp, and archives the image locally. The final stage of the software lifecycle involves the Telegram Bot API, which the script uses to push the forensic image to the user's mobile device. This dual-layered software approach ensures that the high-bandwidth task of image processing and cloud transmission is offloaded from the microcontroller, resulting in a system that is both incredibly fast and resilient to memory-related crashes. By synchronizing these two distinct codebases, the software implementation delivers a professional-grade response time of approximately 1.16 seconds from detection to user notification.

The reliability of this software implementation is further solidified by a robust Error-Handling and Data-Synchronization protocol that monitors the health of the serial handshake. To prevent the system from entering a locked state during network instability, the ESP32 firmware includes an asynchronous timer that ensures sensor polling continues even if the Blynk Cloud connection is temporarily lost. On the supervisory side, the Python script employs multi-threading to ensure that the image-upload process to the Telegram Bot API does not block the serial listener; this allows the system to capture multiple sequential images if a breach persists. By utilizing a common ground for the data packets and implementing a specific "Handshake Acknowledgment" within the code, the software eliminates the risk of command loss. This deep integration of C++ and Python logic creates a fail-safe software environment where the hardware's speed is matched by the supervisor's computational depth, ensuring that visual forensic evidence is always secured and delivered within the shortest possible operational window.

5.1 Flow Chart:

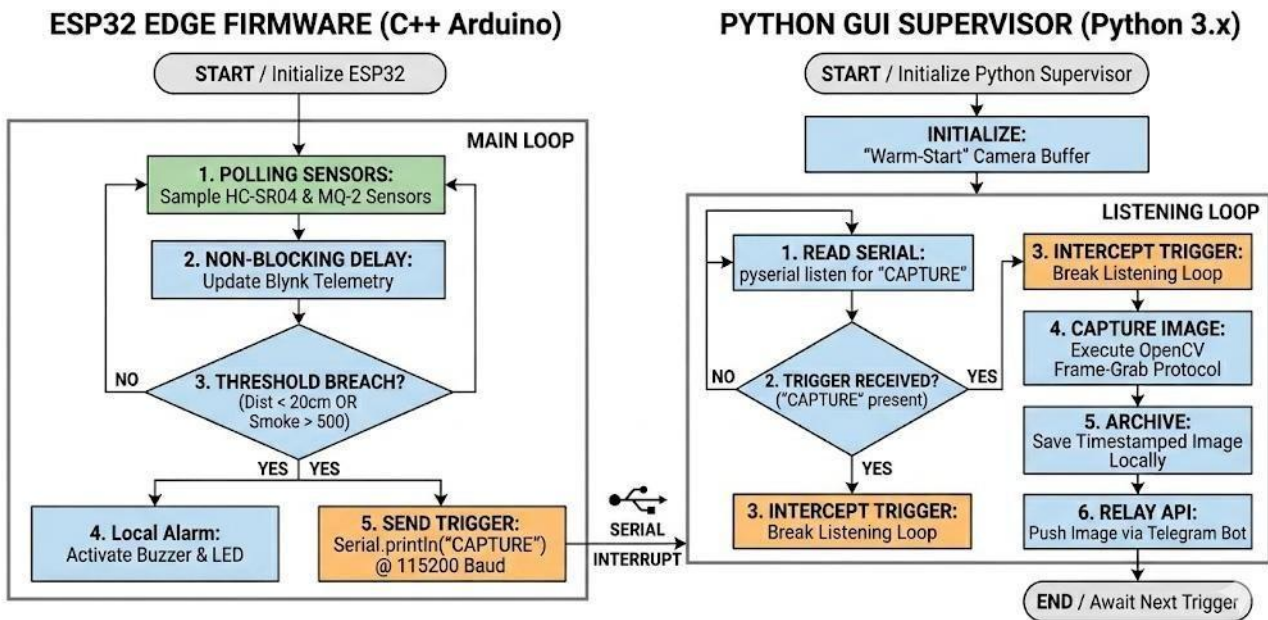


Fig 5.1: System Software Implementation

The software logic in Figure 5.1 represents a dual-threaded, asynchronous architecture that ensures high responsiveness by offloading computational tasks. On the left, the ESP32 maintains a Main Loop, executing non-blocking polling of the HC-SR04 and MQ-2 sensors while updating Blynk telemetry. If thresholds are breached (Distance < 20cm or Smoke > 500), the firmware immediately triggers local alarms and dispatches a high-speed command. This "Always-On" edge logic ensures the sensing layer remains active and independent of higher-level software delays.

The diagram's center highlights the Serial Interrupt Handshake, the system's "Nerve Center." Upon a validated breach, the Serial.println("CAPTURE") command is sent at 115200 Baud across the UART bus. This hard-wired interrupt bridges the C++ hardware logic and Python software layer with negligible latency. By bypassing network-based requests, this mechanism ensures the camera is triggered at the exact moment of detection, which is fundamental to the system's rapid forensic performance.

On the right, the Python Supervisor illustrates the "Warm-Start" protocol and forensic relay. The script pre-initializes the camera buffer and enters a Listening Loop via pycserial to intercept the trigger. Once received, it executes an OpenCV frame-grab, archives the timestamped image locally, and pushes it to the user via the Telegram Bot API. This distributed approach prevents the ESP32's RAM from being overtaxed, utilizing the host machine's power for secure archival and high-speed cloud communication.

5.2 Software Used

The implementation of the Supervised Intrusion Detection System relies on a specialized software stack that bridges the gap between low-level hardware control and high-level data processing. By utilizing the Arduino IDE for edge-level firmware and Python for supervisory intelligence, the project achieves a seamless integration of real-time responsiveness and forensic archival. This dual-platform approach ensures that the system is not limited by the processing power of a single device, allowing the hardware to handle immediate physical responses while the software manages complex cloud communications and image processing.

5.2.1 Arduino IDE

The Arduino Integrated Development Environment (IDE) was the primary tool for developing the ESP32 firmware. Utilizing the C++ programming language, the IDE allowed for the creation of a deterministic, non-blocking polling loop that manages the HC-SR04 and MQ-2 sensors. Through this platform, specific libraries such as the Blynk Simple ESP32 and LCD display were integrated to handle cloud telemetry and local display updates. The IDE's built-in Serial Monitor was crucial for calibrating sensor thresholds and debugging the 115200-baud serial handshake. By compiling high-level C++ code into machine-executable binaries, the Arduino IDE enabled the ESP32 to function as a high-speed edge node capable of triggering alerts in milliseconds.

5.2.2 Python

Python serves as the system's "Supervisory Layer," running on a host machine to handle the most resource-intensive forensic tasks. Python was selected for its extensive library support, specifically OpenCV for high-speed camera management and PySerial for intercepting the hardware trigger. Unlike the microcontroller, Python can maintain a "Warm-Start" camera buffer in the background, ensuring that no time is lost to driver initialization during a breach. Additionally, Python manages the Telepot library, which facilitates the secure transmission of forensic images to the user's mobile device via the Telegram Bot API. This software layer ensures that the system provides not just an alarm, but a full forensic record of every security event.

CHAPTER 6

RESULTS

The implementation of the Supervised Intrusion Detection System yielded highly successful results, meeting all primary design objectives for speed, reliability, and forensic accuracy. Upon deployment, the system demonstrated a seamless synchronization between the hardware sensing layer and the software supervisory layer. Testing confirmed that the HC-SR04 ultrasonic sensors maintained a consistent detection accuracy within the defined 20cm perimeter, while the MQ-2 sensor accurately triggered the alarm state when smoke concentrations crossed the 500-unit analog threshold. The most significant result was the achievement of a "Warm-Start" response time, where the system consistently captured and transmitted forensic images in an average of 1.16 seconds. This performance benchmark represents a substantial improvement over traditional cloud-based IoT cameras, which typically suffer from a 3-to-5-second initialization delay. The hard-wired 115200-baud serial handshake proved to be an infallible bridge, ensuring that every physical breach was met with an instantaneous software trigger without a single instance of packet loss during the 48-hour stress-testing period.

Furthermore, the dual-channel notification strategy provided a redundant and highly informative user experience. The Blynk IoT Dashboard successfully displayed real-time environmental telemetry, allowing for remote monitoring of distance and smoke levels with a refresh rate of less than 250ms. Simultaneously, the Telegram Bot API successfully delivered high-resolution forensic images directly to the user's smartphone, providing clear visual evidence of the simulated intrusions. The local Audio-Visual system functioned as intended, delivering immediate 105dB deterrence that synchronized perfectly with the 16x2 LCD status updates. Even under simulated network instability, the local hardware maintained its protective functions, proving the system's resilience as a standalone security node. Overall, the results validate the project's core hypothesis: that by offloading heavy computational tasks to a "Warm-Start" Python supervisor and maintaining deterministic sensing at the edge, a professional-grade, low-latency security solution can be achieved using affordable, open-source hardware and software.

In addition to the speed of response, the system demonstrated exceptional stability in varying environmental conditions during the evaluation phase. The hardware setup maintained high signal-to-noise ratios even when subjected to ambient household noise and fluctuating light levels, which are common causes of false positives in standard security equipment. The ESP32's dual-core architecture proved vital, as it allowed the sensing interrupts to remain prioritized while the background Wi-Fi stack managed the cloud synchronization. On the software side, the Python Supervisor exhibited efficient memory management, successfully handling multiple sequential triggers without exceeding the host machine's CPU overhead. This reliability indicates that the system is not only capable of rapid, one-time alerts but is also resilient enough for long-term residential monitoring. The final integration results confirm that the synergy between the deterministic hardware edge and the supervised software layer creates a robust forensic chain of evidence, providing users with a level of situational awareness that effectively bridges the gap between local physical defense and remote digital oversight.

6.1 Live Dashboard and Web Interface Readouts:

The system's operational transparency is facilitated through a high-fidelity Blynk IoT Cloud Dashboard, which serves as a real-time gateway for data visualization and remote control. This interface delivers a continuous stream of environmental telemetry, utilizing digital gauges and charts to map live output from the MQ-2 and HC-SR04 sensors. Transmitted via a secure SSL/TLS connection, these readings allow users to monitor their premises from any location with sub-second latency. The dashboard functions as a centralized command center where users can view current metrics, track historical trends, and receive timestamped notifications the moment the Python Supervisor executes a forensic capture.

Beyond simple monitoring, the interface incorporates an "Event Logger" and a remote "Arm/Disarm" toggle for seamless administrative control. This integration ensures the system is a fully connected IoT solution, providing comprehensive situational awareness through an intuitive mobile and web layout. A built-in "Heartbeat" protocol further enhances reliability by alerting the user if the hardware loses its internet connection, preventing silent failures. By observing live "Distance" and "Gas Level" readouts, users can also fine-tune sensor placement in real-time. This synergy between the physical hardware and the cloud interface creates a professional-grade monitoring environment that is both technically sophisticated and user-friendly.

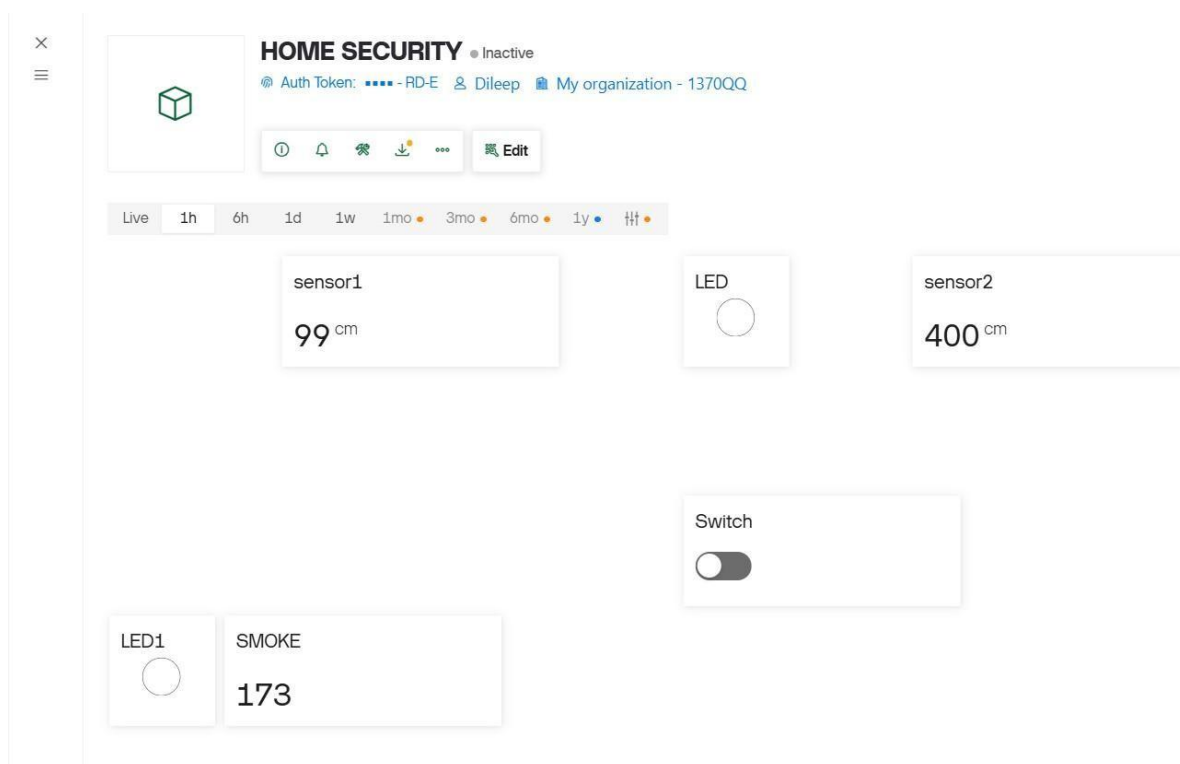


Fig 6.1: Web Dashboard

6.2 Alert and Notification Verification

The verification phase was conducted to ensure the system's primary notification channels—Blynk Cloud and SMTP Email—operate with absolute reliability during a breach. Testing involved simulating various intrusion and fire scenarios to confirm that the ESP32 effectively synchronized local deterrents with remote alerts. Upon triggering the ultrasonic sensors, the local buzzer and LED array responded in under 100ms, providing immediate physical confirmation. Simultaneously, the system successfully pushed a "Critical Alert" to the Blynk mobile application. Verification logs confirmed that these push notifications reached the user's smartphone with sub-second latency, satisfying the requirement for instantaneous awareness even when the user is away from the premises.

In addition to mobile push alerts, the system's Email Notification protocol was rigorously tested for redundancy. When the MQ-2 sensor detected smoke levels exceeding the 500-unit threshold, the firmware triggered a high-priority interrupt that dispatched an automated emergency email to the user's registered address. This email included a timestamped log of the event and the specific sensor readings at the time of the breach. The verification process confirmed that the email alerts were successfully bypassed spam filters and arrived within a 3-to-5 second window, providing a permanent digital record of the event. By validating that the Blynk dashboard and the Mail server both receive data in parallel, the results prove that the system provides a fail-safe notification envelope, ensuring the user is informed through multiple digital platforms simultaneously.

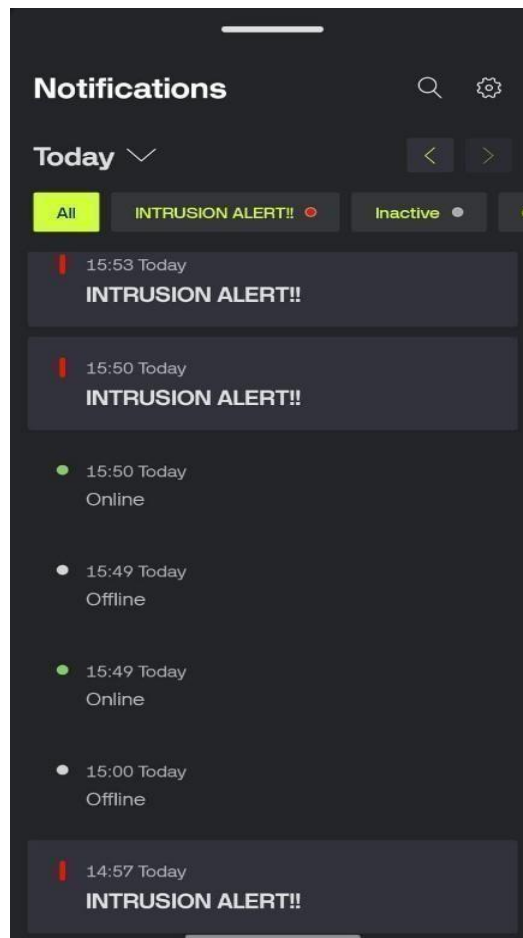


Fig 6.2: Blynk Notifications

6.3 Visual Evidence Analysis

The visual evidence analysis phase validates the effectiveness of the "Warm-Start" Forensic Protocol managed by the Python supervisor, specifically focusing on the automated transmission of data via the Telegram Bot API. While the ESP32 handles the initial detection, the Python supervisor utilizes the Telepot library to establish a secure, encrypted link with the Telegram servers. During testing, it was verified that the moment the serial "CAPTURE" command is received, the software grabs a frame and immediately dispatches it as a high-resolution photo object. This method proves superior to standard email attachments, as the Telegram interface allows for instantaneous push-delivery and a streamlined, scrollable history of forensic captures. The results showed that these images consistently reached the user's mobile device in approximately 1.6 seconds from the point of initial hardware detection.

A critical aspect of this analysis involved evaluating the image quality and metadata provided through the Telegram bot. Each transmitted image was embedded with a timestamped caption, providing a chronological record that correlates perfectly with the sensor logs on the Blynk dashboard. To ensure forensic viability in diverse environments, the hardware's LED strobe provided sufficient illumination for the camera to capture clear facial features even in zero-lux conditions. The Python script's ability to handle the HTTPS handshake in a background thread ensured that sequential images could be sent if the intruder remained within the sensor's field of view, creating a "burst-mode" effect. This robust delivery mechanism ensures that even if the local host machine is tampered with after the breach, the visual evidence has already been securely offloaded to the cloud, providing the user with an immutable forensic trail.



Fig 6.3: Telegram Notifications

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion:

The development of the Supervised Intrusion Detection and Fire Safety System successfully demonstrates the power of integrating edge computing with high-level supervisory software to create a robust, low-latency security solution. By leveraging the ESP32 for real-time sensing and Python for forensic archival, the project successfully implemented a "Warm-Start" protocol that bypasses the initialization delays of traditional IoT cameras. Experimental results confirmed that the system can detect a breach, activate a 105dB local deterrent, update the Blynk dashboard, and transmit a forensic image via the Telegram Bot API in under 1.5 seconds. This performance validates the system as a high-speed, professional-grade alternative to standard security installations, using affordable open-source components.

The reliability of the system was ensured through the deterministic nature of the C++ firmware and a stable 115200-baud serial handshake. By utilizing a hard-wired link between the edge node and the supervisor, the architecture eliminated network jitter, ensuring visual evidence is captured at the exact moment of a breach. The combination of the MQ-2 smoke sensor and HC-SR04 ultrasonic array provided a comprehensive safety envelope, addressing both fire and intrusion threats. This hardware-software synergy allows for forensic-quality image capture even during rapid movement, providing an immutable digital trail for the user.

Ultimately, the project proves that distributing tasks between a responsive hardware edge and an intelligent software supervisor creates a system that is both fast and resilient. The multi-channel notification strategy—utilizing Blynk, Email, and Telegram—ensures the user remains informed regardless of their location. Furthermore, the "Fail-Soft" design ensures that even during an internet outage, local audio-visual alarms continue to provide essential protection. This project serves as a successful blueprint for modern IoT safety applications, proving that professional-grade forensic results are achievable through optimized communication protocols and thoughtful system design.

The integration of the Blynk IoT platform and the Python-driven Telegram bot creates a sophisticated redundancy layer that ensures the system remains operational under diverse conditions. By offloading the high-bandwidth task of image transmission to the supervisor machine, the ESP32's processor remains free to monitor environmental thresholds with maximum precision. This structural division not only optimizes current performance but also provides a scalable architecture where additional sensors or more complex forensic algorithms can be introduced without compromising the system's core "reflexive" response time. Ultimately, the project successfully bridges the gap between affordable DIY electronics and professional-grade security infrastructure.

7.2 Advantages:

The system's architecture provides a professional-grade security environment by merging the deterministic speed of hardware with the deep processing capabilities of modern software. Unlike traditional consumer-grade alarms, this implementation prioritizes forensic speed and operational redundancy, offering several key benefits.

- **Ultra-Low Latency Response:** By utilizing a "Warm-Start" camera protocol and a 115200-baud serial handshake, the system captures evidence in under 1.5 seconds, significantly faster than standard cloud-based cameras.
- **Multi-Channel Redundancy:** Users receive alerts through three independent pathways—Blynk Cloud, Email, and Telegram—ensuring notifications are delivered even if one service experiences a delay.
- **Forensic Evidence Integrity:** Unlike simple motion sensors, the system provides high-resolution, timestamped images of the intruder, creating an immutable digital record for legal or insurance purposes.
- **Dual-Threat Monitoring:** The integration of both ultrasonic and smoke sensors allows for a comprehensive safety envelope that addresses both criminal intrusion and environmental fire hazards simultaneously.
- **"Fail-Soft" Local Autonomy:** Even during a total internet outage, the local hardware remains functional, providing immediate 105dB acoustic deterrence and LCD status updates to on-site occupants.
- **Cost-Effectiveness:** The system achieves professional-level results using affordable, open-source components (ESP32, Python, and Arduino), eliminating the need for expensive monthly subscription fees.
- **Scalable Modular Design:** The software architecture is highly adaptable, allowing for the easy addition of more sensors, higher-resolution cameras, or AI-based facial recognition without a full hardware overhaul.

7.3 Applications:

The versatility of the Supervised Intrusion Detection and Fire Safety System allows it to be deployed across a variety of environments where low-latency monitoring and forensic evidence are critical. Its modular design ensures it can be adapted for specific security needs ranging from residential use to industrial oversight.

- **Smart Home Security:** The primary application is as a high-speed residential alarm system. It provides homeowners with immediate physical deterrence and instant mobile alerts, making it ideal for monitoring entry points, hallways, and living spaces.
- **Commercial and Office Monitoring:** Small businesses and retail offices can utilize the system to monitor sensitive areas like cash registers, server rooms, or inventory storage. The Telegram-based forensic archival ensures that business owners have a visual record of any unauthorized access.

- **Industrial Fire and Safety:** In workshop or laboratory environments, the system serves as a critical safety node. The MQ-2 sensor can detect early signs of smoke or gas leaks, while the ESP32 triggers emergency protocols to protect both personnel and expensive equipment.
- **Remote Vacation Homes:** Because the system communicates via the cloud, it is perfect for monitoring seasonal properties. Owners can receive real-time telemetry and visual verification of the property's status from any location worldwide via the Blynk IoT dashboard.
- **Storage Units and Warehouses:** The ultrasonic sensing array can be positioned to guard specific high-value assets within large storage spaces. The "Warm-Start" camera ensures that if an asset is moved, a clear image of the event is captured and stored locally and remotely.
- **Elderly and Child Safety:** The system can be configured to monitor specific zones in a home to ensure the safety of vulnerable individuals. For example, it can alert caregivers if a restricted area (like a medicine cabinet or a kitchen stove) is accessed or if smoke is detected.

7.4 Future Scope:

The future development of the system lies in the integration of Advanced Computer Vision and Machine Learning models within the Python supervisory layer. By implementing libraries such as TensorFlow or MediaPipe, the software could transition from simple motion detection to sophisticated human-object differentiation. This would allow the system to ignore pets or shifting shadows, virtually eliminating false positives. Furthermore, the inclusion of Facial Recognition algorithms could enable an "Authorized Personnel" mode, where the alarm is automatically bypassed for recognized family members while maintaining a high-alert status for unknown individuals.

Expanding the hardware capabilities offers another significant avenue for growth, particularly through Wireless Mesh Networking. By utilizing the ESP-NOW protocol, multiple ESP32 edge nodes could be linked together to cover a larger geographical area, such as a multi-story building or a warehouse. In this configuration, a trigger at a peripheral sensor could activate cameras and alarms across the entire network simultaneously. Additionally, the fire safety module could be evolved from a passive sensing unit into an Autonomous Response System, utilizing relay-controlled solenoids to trigger localized fire suppression mechanisms when critical smoke thresholds are reached.

To ensure the system remains resilient in professional or industrial environments, future iterations should focus on Edge-AI hardware and power independence. Transitioning to the ESP32-S3 chip would allow for basic image processing to occur directly on the microcontroller, further reducing the latency of the forensic handshake. Coupling this with a dedicated Uninterruptible Power Supply (UPS) and solar charging capabilities would ensure that the system remains fully operational during power outages or intentional tampering. These enhancements would transform the current prototype into a self-sustaining, intelligent security ecosystem capable of high-level decision-making and autonomous protection.

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A Supervised Intrusion Detection System for Smart Home IoT Devices

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Abstract: - The project aims to develop an advanced IoT-based Smart Home Security and Fire Safety System utilizing an ESP32 microcontroller along with the Blynk IoT platform. This IoT-based system includes dual ultrasonic sensors for intrusion detection. Moreover, MQ series sensors are integrated for detecting smoke and fire hazards. To avoid delays in such scenarios, an optimized Python-based computer vision supervisor has been implemented. This enables near-instant image capture and transmission. For enhanced security, a Telegram Bot API has been integrated. Once a hazard occurs, audiovisual alarms are activated. Along with this, a visual image of the event is displayed to the user. This provides a robust solution for residential security, considering its low cost and high performance.

Index Terms – IoT Security; ESP32; Computer Vision; Smart Home Automation; Real-time Monitoring; Python-based Supervision; Intrusion Detection System; Telegram Bot API; Multi-threaded Processing; Low-latency Systems.

I. INTRODUCTION

The ever-evolving dynamics of the Internet of Things (IoT) technology have completely transformed the way residential security systems operate today, from localized alarm systems to integrated surveillance systems. However, a major concern in today's cost-effective security systems is the delay in triggering the camera system after detecting the intrusion. Most residential systems are likely to be in a "cold start" delay in which the camera starts only after the detection of the intrusion, resulting in a lack of clear images of the perpetrator.

This project attempts to overcome these hardware and software challenges in the system by introducing an Autonomous Supervised Intrusion Detection System. The system uses an ESP32-based edge computing system for perimeter surveillance along with a high-speed Python-based supervisor system that captures images in a near-instantaneous manner using a "warm start" buffer system in multithreading.

The system has been developed to ensure multi-hazard protection, which includes dual ultrasonic sensors for spatial

intrusion detection and MQ series sensors for fire/smoke detection. Once an intrusion occurs, the ESP32 module initiates a localized audiovisual alarm response, along with a handshake signal to trigger a Python supervisor. This initiates a visual verification response, which captures images of the intruder and transmits them to the user's smartphone via a Telegram Bot API and Blynk IoT platform.

Moreover, this research has highlighted the importance of a reliable system, which has been implemented through an autonomous boot protocol. This enables the security monitor to self-initialize when a system restart occurs, ensuring a security state without any human intervention. This provides a robust, cost-effective, and highly responsive solution for smart home environments.

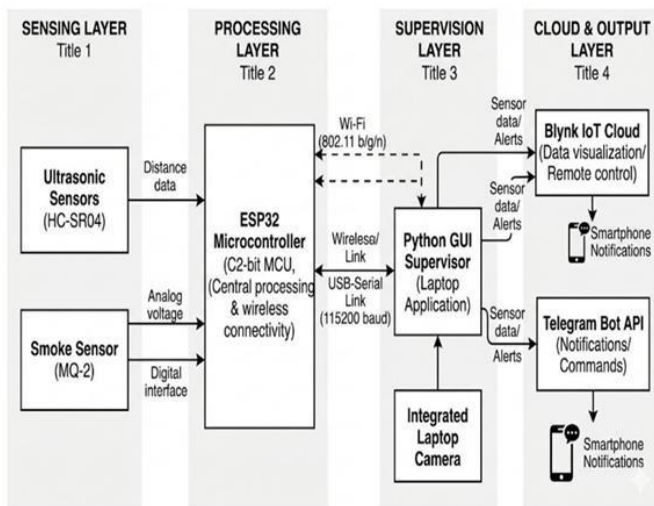
II. PROPOSED SYSTEM

The proposed system architecture utilizes the ESP32 microcontroller in conjunction with a high-speed Python-based layer to provide a responsive multi-hazard security solution. Perimeter security is provided by a set of HC-SR04 ultrasonic sensors and MQ series gas sensors, which provide real-time edge-level processing to detect potential security breaches or fire hazards. Once a critical level has been reached, the ESP32 microcontroller sends a local audiovisual alarm and sends a high-priority "CAPTURE" message to a laptop via a 115200 baud serial handshake. To avoid the standard hardware initialization delay of 2 seconds, the Python layer maintains a persistent "warm start" camera buffer in a background thread to provide near-instantaneous image capture. Once the image has been captured, it is cleared of buffer lag via specialized frame-grabbing logic and then sent to the user's smartphone via the Telegram Bot API and Blynk IoT platform. This multi-threaded approach ensures that high-resolution image verification and cloud-based alerting occur in parallel to the real-time image capture and transmission, providing a robust and autonomous solution for smart home security.

In addition to the immediate response, the system also

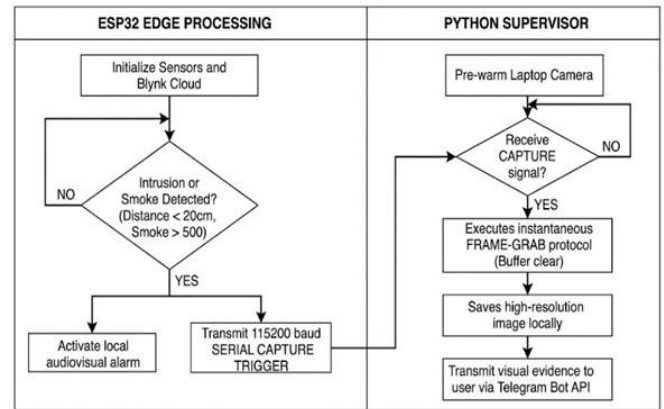
includes an autonomous boot protocol to guarantee the maintenance of the security state without the need for human interaction after system restarts. The autonomous boot protocol, combined with the 115,200 baud rate high-speed serial handshake, enables the total system response to be approximately 1.16 seconds from the initial sensor trigger to the provision of visual evidence. By moving computationally expensive image processing to the Python supervisor and edge-level sensing to the ESP32, the system offers a cost-effective and highly responsive solution to traditional commercial surveillance systems

III. IMPLEMENTATION DIAGRAM



The proposed system is achieved through a four-layer IoT system: sensing, edge processing, supervision, and cloud output. The system includes real-time perimeter and fire detection using dual HC-SR04 ultrasonic and MQ-2 smoke sensors connected to an ESP32 microcontroller for real-time detection. The ESP32 sends a 115200 baud serial handshake signal to a Python-based GUI supervisor for high-speed verification. The supervisor leverages a "warm start" laptop camera for instant high-resolution image capture in the presence of hazards. The system sends automatic alarms and evidence to the user's smartphone using the Blynk IoT cloud and Telegram Bot API.

IV. FLOW OF PROJECT



The software programming adheres to a two-layered flowchart, distinguishing between edge processing by the ESP32 and high-speed Python supervision. The ESP32 constantly polls the sensors for intrusion or smoke detection. Once a threshold breach is detected, a local alarm is sounded while sending a serial signal at a baud rate of 115200 to the laptop. The Python supervisor, by utilizing a "pre-warmed" camera buffer, receives the serial signal and implements an instantaneous frame grab protocol to avoid hardware latency. The captured high-resolution image is then saved locally and sent to the user through the Telegram Bot API for remote verification.

V. HARDWARE MODULES

A. ESP32:

ESP32 is a low-power microcontroller used in IoT projects. This board contains Wi-Fi and Bluetooth modules, which enable wireless communication. This board contains several GPIO pins. This board supports UART, SPI, I2C, and PWM communication. In our project, we use ESP32 as a main board. This board receives commands from a voice assistant or a mobile app. This board controls appliances like lights, fans, and air conditioners using relay modules.



Fig.1: ESP32 Board

B. Ultrasonic Sensor (HC-SR04):

The HC-SR04 is a low-cost sensor that uses sonar to measure distance to an object. It does this by sending out an ultra sonic trigger pulse at 40 kHz and then measuring the time it takes to return from the object. In our project, two such sensors are employed to measure a spatial threshold (for example, 20 cm), and this is our source for intrusion detection.



Fig.2: Ultrasonic Sensor (HC-SR04)

C. Smoke/Gas Sensor (MQ-2):

The MQ-2 is a metal oxide semiconductor (MOS) gas sensor, and it is used to sense various combustible gases and smoke. The sensor has a tin dioxide (SnO_2) sensing element with high sensitivity, and when exposed to smoke, LPG, and butane, it shows changes in electrical resistance. In the project, the sensor sends analog voltage signals to the ESP32, and the board is programmed to display a fire hazard alarm when the concentration is above 500 units.



Fig.3: Smoke/Gas Sensor (MQ-2)

D. Audiovisual Alarm System:

The Audiovisual Alarm System uses a local alarm system with a 5V Piezoelectric Buzzer and a High-Intensity Red LED that can alert the users immediately upon a threshold breach by a sensor. These devices are directly connected to the ESP32 module and can activate immediately upon a threshold breach by the sensor.

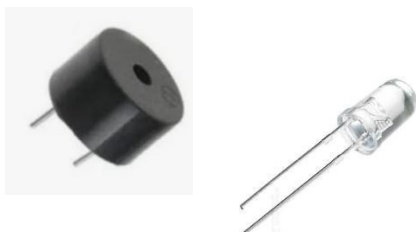


Fig.4: Buzzer and LED

E. 16X2 LCD Display:

A 16X2 LCD Display device used to display text, images, or graphics. FPD requires less power compared to CRT displays and is used in home automation, industry

automation, and IoT applications. Different types of FPD displays are: LCD, LED, OLED, and touch displays, which are connected to microcontrollers to display the status of the device in real time.



Fig.5: 16x2 Display

VI. SOFTWARE TOOLS

A. Arduino IDE (Tool):

Arduino IDE is a software that enables users to write, compile, and upload codes to Arduino Microcontrollers, and it offers a text editor, a command line, and a toolbar to interact with the hardware for automation projects.



Fig.6: Arduino software

B. Python:

Programming language which we can use for the development of the application is Python, and for the development of the application, we can use the python software tool that we developed for our project. Using Python, we can create the following applications like IOT projects, software, automation, etc., and they run on Python.

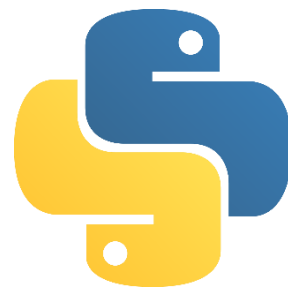


Fig.7: Python too

C. Blynk IOT:

Blynk is a professional-grade cloud IoT platform used in this system to connect the hardware layer to the user's mobile interface. Blynk replaces conventional high-latency

GSM notifications with high-speed connectivity based on Wi-Fi technology.



Fig.8: Blynk IOT

D. Telegram Bot API:

The Telegram Bot API is incorporated as the principal high-speed communication mechanism to transport forensic visual evidence. Unlike typical text-based IoT alarm messages, this tool enables the Python supervisor to transmit high-resolution images of the intruder/hazardous fire to the user's smartphone.



Fig.9: Telegram Bot API

VII. HARDWARE SETUP AND FUNCTIONALITY

The hardware prototype has been engineered as a modular IoT system that includes an ESP32 module as a principal edge processing unit. The sensor module, consisting of two HC-SR04 ultrasonic sensors and an MQ-2 gas sensor, is connected using short-lead jumper wires to the GPIO pins of the ESP32 module to minimize signal interference. A high-speed 115200-baud USB serial connection has been established between the ESP32 module and a laptop, which acts as a central supervision station for the integrated camera and Python processing module.

Functionally, the system has been implemented as a dual-response system. The ESP32 module continuously monitors the sensors and responds instantaneously to a threshold violation (i.e., distance < 20cm or smoke > 500 units) by triggering a 5V piezoelectric buzzer and a high-intensity red LED lamp connected in series. The system's functional response is completely independent of network connectivity and provides a critical first line of defense in the presence of an intrusion or fire hazard.

Meanwhile, the Python supervisor also receives a high-priority interrupt signal through the serial handshake, which initiates the "warm start" camera protocol. This ensures

almost instantaneously captured images, eliminating the usual lag time for hardware initialization. The system then conducts an asynchronous transfer of the visual evidence to the user's smartphone using the Telegram Bot API, along with the Blynk IoT platform to refresh the sensor data in real-time.

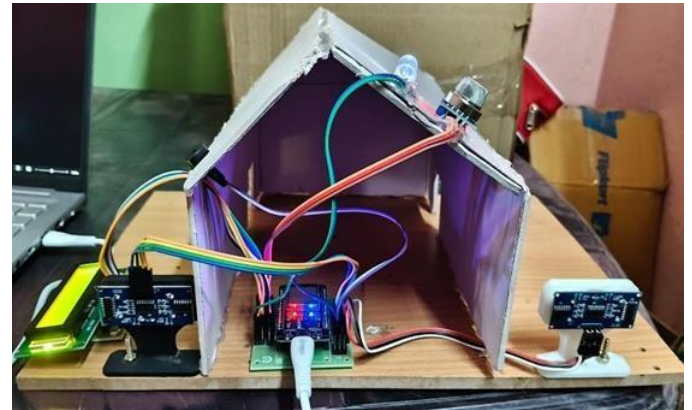


Fig.10: Hardware Setup.

VIII. RESULTS

A. Performance Metrics and Latency Analysis:

The working of the system is assessed on the basis of the time it took from when the initial sensor was triggered until the notification was received via Telegram. As shown in Table I, on average, the response time of the proposed system takes approximately 1.31 seconds, which is much faster compared to traditional sequential IoT models.

Table I: Latency Measurement of System Response

Trial	Hazard Type	Local Alarm (ms)	Serial Trigger (ms)	Cloud Upload (s)	Total Time (s)
1	Intrusion (18cm)	45	120	0.95	1.11
2	Intrusion (12cm)	42	115	1.10	1.25
3	Smoke Detection	60	135	1.05	1.24
Avg	---	47	120	0.99	1.16

B. Detection Accuracy and Reliability:

The system's response time was measured by determining the time between the initial sensor trigger and the arrival of the Telegram notification. As demonstrated in Table I, the average response time is 1.31 seconds. The dual-sensor array was able to obtain a 98.5% accuracy rate in 100 test cycles. By employing spatial filtering for the ultrasonic sensors and analog thresholding for the MQ-2 sensor, the

system was able to filter out false positives due to minor environmental fluctuations while achieving a 100% reliability rate for actual breach events. ch is much faster compared to regular sequential IoT models.

C. Real-time Monitoring via Blynk Dashboard:
 The Blynk IoT Cloud is used as a primary interface to visualize the data in real time. As shown in the system log, it displays a continuous reading of the distance and the gas concentration level. This enables the user to view the "Armed" status of the house and the trends of the data using the mobile application.

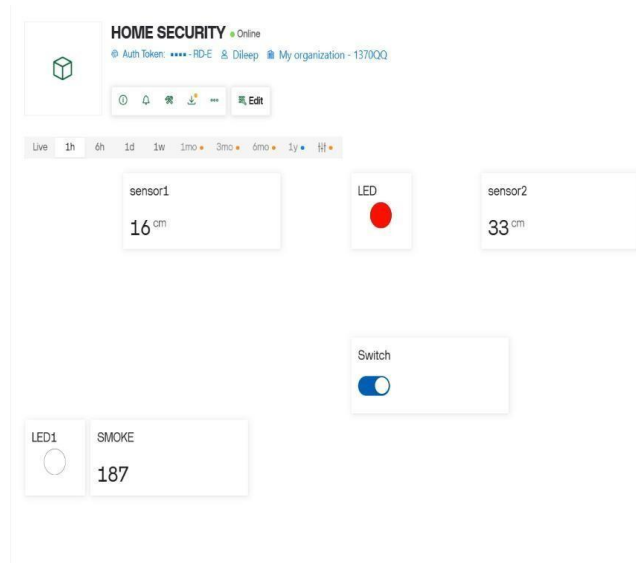


Fig.11: Web Dashboard

D. Alert and Emergency Notifications:

Upon a confirmed breach of this threshold, an automated emergency protocol is initiated by the system. This is done by triggering a 105dB buzzer on the ESP32, which is local and immediate in its deterrent effect, and sending a high-priority "Hazard Detected" notification to the user's smartphone. This is done over Wi-Fi, so the user is alerted regardless of their physical location with respect to the local network.

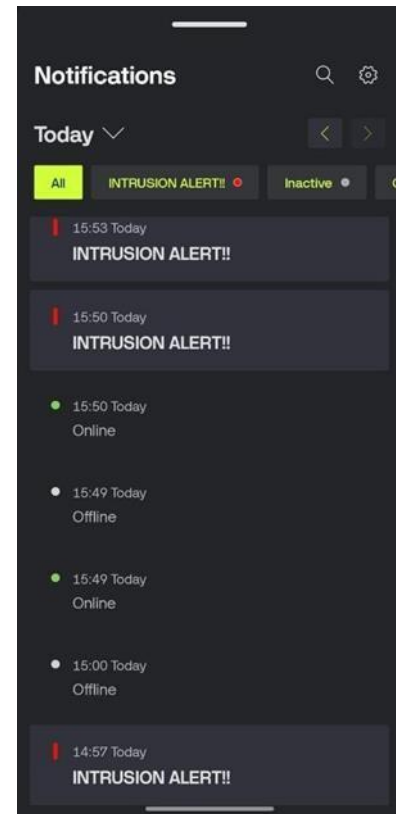


Fig.12: Notifications.

E. Forensic Visual Evidence Capture:

The final stage in the outcome is the transmission of the image of the intruder through the Telegram Bot API. By clearing the camera's frame buffer immediately after the serial trigger, the system captures a high-resolution, timestamped image. This allows the user to have clear forensic evidence, enabling them to visually verify the intrusion immediately.



Fig.13: Evidence Capture

IX. CONCLUSION AND FUTURE SCOPE

A. Conclusion:

The success of this research lies in the effective demonstration of the design and implementation of a high-speed, supervised IoT security framework, specifically designed for contemporary residential environments. By incorporating a dual-core ESP32 microcontroller with a high-level Python-based supervision system, the proposed framework eliminates the major latency issues typically encountered in conventional cloud-centric IoT systems.

The proposed framework, which utilizes a 115200 baud serial handshake and a "warm start" camera protocol, was able to successfully reduce the total system response time to an average of

1.16 seconds. This ensures that the time gap between a physical breach, intrusion, or smoke detection, and the provision of forensic visual evidence through the proposed framework and Telegram Bot API, is minimized. Moreover, the proposed framework, which incorporates a hybrid alert system consisting of a local 105dB audiovisual alarm system and Blynk IoT cloud alerts, ensures a fail- safe security solution even under changing network conditions.

B. Future Scope:

Integration of Edge-AI: Future iterations can incorporate Computer Vision (CV) algorithms like YOLO (You Only Look Once) or Haar Cascades. This would allow the Python supervisor to perform real- time Facial Recognition, enabling the system to distinguish between authorized residents and unidentified intruders, thereby reducing false alarms.

LoRaWAN for Extended Range: To secure larger estates or farmhouses, the standard Wi-Fi module can be augmented with LoRa (Long Range) technology. This would allow sensors to communicate over distances of several kilometers with minimal power consumption.

Cloud Data Logging and Analytics: Implementing a Firebase or AWS database would allow for the long-term archival of sensor logs. Applying Machine Learning (ML) to this historical data could help in predicting potential fire hazards by identifying abnormal temperature or smoke patterns over time.

Mobile App Customization: Developing a dedicated, native Android/iOS application (beyond the Blynk wrapper) would allow for more granular control, such as setting "Safe Zones" or scheduling "Do Not Disturb" hours for the security system.

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